Skin biology for the impatient

Week 3 Lecture 1 - Looksmaxxing

Autumn 2026

What we'll cover

- The skin barrier: what it is and what damages it
- Sebum, the microbiome, and why most acne starts at the follicle
- Photoaging: cumulative UV damage and collagen loss
- Cell turnover: why speed matters for appearance
- Retinoids: the mechanism and why they are the top evidence-backed intervention after sunscreen

The skin barrier

The stratum corneum is not dead skin to scrub away

- The outermost layer of skin: flattened dead cells (corneocytes) held together by a lipid matrix
- The lipid matrix contains ceramides, cholesterol, and fatty acids in a specific ratio
- This matrix prevents water loss and blocks irritants and pathogens from entering
- A damaged barrier lets irritants in and water out: redness, stinging, sensitivity, and flaking are all signs

Dr. Dray (2023): "When trying to heal the skin barrier, lay off exfoliants and consider decreasing cleansing frequency to once a day."

What damages the barrier

- Over-cleansing: surfactants that strip lipids raise skin pH and disrupt the lipid matrix
- Physical scrubs and over-exfoliation: direct mechanical disruption of the stratum corneum
- Low humidity and cold air: accelerate transepidermal water loss (TEWL)
- Over-applying active ingredients too fast: retinoids, AHAs, and BHAs all thin the upper layers if not introduced gradually
- Sun exposure: UV radiation degrades structural proteins and lipids

Ceramides, niacinamide, and panthenol support barrier repair

- Ceramides replace depleted lipids in the stratum corneum
- Niacinamide (vitamin B3) stimulates ceramide production and reduces inflammation
- Hyaluronic acid draws water into the epidermis, reducing TEWL
- Panthenol (pro-vitamin B5) promotes cell differentiation and wound healing

These ingredients appear in basic drugstore moisturizers, not just expensive products. The claim that price tracks efficacy is not supported for barrier repair.

Sebum, acne, and the follicle

Acne starts at the follicle, not on the surface

- Sebaceous glands secrete sebum (a lipid mixture) through the hair follicle onto the skin surface
- When a follicle becomes blocked: dead cells and sebum accumulate, forming a comedone
- Cutibacterium acnes (C. acnes) bacteria colonize the blocked follicle and trigger inflammation
- Result: papules, pustules, or cysts depending on depth and immune response

Over-washing does not cure acne. It strips surface lipids, triggers compensatory sebum production, and worsens the barrier without clearing the blockage.

Skin type is not permanent

- Oily, dry, and combination are descriptions of current sebum output, not fixed categories
- Sebum production changes with age, hormones, climate, and the products you use
- A teenage boy with oily skin may have normal-to-dry skin in his late 20s
- Products designed for “oily skin” that over-strip lipids can convert normal skin to reactive skin over months

This matters for routine-building: match products to how your skin behaves now, not to a category you were assigned at 16.

Photoaging

UV exposure is the largest controllable cause of visible skin aging

- UVB (290-320 nm): causes sunburn, drives direct DNA damage, main driver of skin cancer
- UVA (320-400 nm): penetrates deeper, degrades collagen and elastin, drives photoaging
- Photoaging signs: fine lines, uneven tone, loss of firmness, and dyspigmentation are largely UV-driven
- Sun exposure accumulates: a 25-year-old who has never used sunscreen has years of subclinical UV damage that will become visible in the next decade

AAD (2023): "Choose a sunscreen with an SPF of at least 30, broad-spectrum protection against UVA and UVB rays."

Sunscreen is the only anti-aging intervention with consistent evidence

- Most “anti-aging” claims in skincare are not well-supported. Vitamin C serums, peptide creams, and collagen supplements all have limited or mixed evidence.
- Sunscreen has demonstrated efficacy for preventing new photoaging in multiple RCTs
- Retinoids have demonstrated efficacy for reversing some existing photoaging
- Every other topical anti-aging category lacks that level of evidence

Be skeptical of product marketing that cites “clinical studies” without naming the study design, sample size, or comparator.

Cell turnover and retinoids

Cell turnover determines how quickly skin refreshes

- The epidermis replaces itself completely every 28-40 days in young adults
- Turnover slows with age: the same cycle takes 40-60+ days by the mid-40s
- Slow turnover means dead cells accumulate longer, pores appear larger, and dullness increases
- Faster turnover reduces the window during which UV-damaged cells remain at the surface

Cell turnover is not something you can directly observe, but the products that affect it have measurable surface effects.

Retinoids: mechanism and evidence

- Retinoids are vitamin A derivatives that bind to nuclear receptors in skin cells (RAR and RXR)
- Effect on turnover: they speed up differentiation, moving cells from the basal layer to the surface faster
- Effect on collagen: they stimulate collagen production and inhibit matrix metalloproteinases (MMPs) that break it down
- Effect on acne: they prevent follicular plugging (comedolytic effect) and reduce inflammatory lesions

Michelle Wong PhD, Lab Muffin (2019): "Retinoids are the next essential product after sunscreen because they have proven efficacy across so many skin issues."

The retinoid ladder

Strongest evidence, requires prescription:

- **Tretinoin** (all-trans retinoic acid): 0.025%, 0.05%, 0.1%; direct receptor binding

Weaker but OTC accessible:

- **Retinol**: converts to retinaldehyde then to retinoic acid in skin; 3-10x less potent per unit
- **Retinal (retinaldehyde)**: one step from retinoic acid; stronger than retinol, less irritating than tretinoin

The conversion steps mean OTC retinoids work but require higher concentrations for equivalent effect. Irritation is lower; the adjustment period is still present.

What retinoids do NOT do: they do not change bone structure; they do not eliminate deep scarring; and they take 12 weeks minimum before any anti-aging effect appears. The purging phase in weeks 1-4 (increased breakouts) is caused by accelerated comedone turnover, not product failure.

Take-aways

1. The skin barrier is a lipid-protein matrix. Disrupting it causes sensitivity, dryness, and increased acne. Repair it before adding actives.
2. Acne starts at the follicle, not on the surface. Over-washing makes it worse by stripping lipids and triggering more sebum production.
3. UV exposure is cumulative and the largest controllable driver of visible aging. Sunscreen is the only anti-aging intervention with consistent RCT evidence.
4. Retinoids speed up cell turnover, stimulate collagen, and are comedolytic. They are the top evidence-backed topical after sunscreen.
5. Skincare does not change bone structure. Most “anti-aging” serum claims are not well-supported.

> "Retinoids are the next essential product after sunscreen because they have proven efficacy across so many skin issues -- and almost every concern worth addressing in a skincare routine responds to retinoids."

-- Michelle Wong PhD (Lab Muffin Beauty Science), *How to Start on Retinoids*, 2019

What's next

- **Week 3 Lecture 2:** The four-product routine: cleansers, sunscreen, moisturizer, retinoids, and acne treatment
- **Week 3 Reading:** Skincare from skin biology forward (required before section)
- **Week 3 Section:** Bathroom shelf audit. Bring your current products or a photo of your shelf.
- **HW1 due this week:** Honest baseline audit